

Respiratory Cilia and Airway Defense: From Physiology to Clinical Disorders

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Abstract

Mucociliary clearance is the hidden backbone of the respiratory system which functions to maintain sterile environment in the airways and also contributes to pulmonary homeostasis, done majorly with the help of specialized motile organelles called as respiratory cilia. Mucus, trapped pathogens, pollutants from the environment and particulate matter all get propelled out by the coordinated beating of cilia toward the upper airway for elimination that is largely dependent on ciliary ultrastructure along with the right thickness, viscosity, elasticity, and flow behavior of the mucus and adequate hydration of the airway surface.

A large spectrum of airway conditions and disorders can result in dysfunction of respiratory cilia including disorders like primary ciliary dyskinesia (PCD), chronic obstructive pulmonary disease (COPD), cystic fibrosis (CF), asthma, bronchiectasis, and chronic rhinosinusitis.

Recent advances in the last few years have substantially contributed in our understanding of ciliary biology, molecular genetics, epithelial signaling pathways, and translational therapeutics. Also, smarter innovations such as high-speed video microscopy, cryoelectron tomography, next-generation genetic sequencing, artificial intelligence-assisted diagnostics, organoid models, and emerging gene therapies have also transformed both research and clinical management of ciliary disorders.

Based on recent literature available in the last few years, this review discusses the basic physiology of respiratory cilia, mechanisms regulating mucociliary clearance, pathophysiology of ciliary dysfunction, recent diagnostic and therapeutic advances, and future perspectives in respiratory ciliary medicine.

Keywords: respiratory cilia, mucociliary clearance, airway defense, primary ciliary dyskinesia, pulmonary physiology, airway epithelium, respiratory disease

Introduction:

The respiratory system is in constant exposure and interaction with the external micro inhabitants of the environment namely, airborne microorganisms, allergens, particulate pollutants, smoke, and environmental toxins. However, maintenance of a sterile pulmonary environment is one of the fundamental requirements of the body so as to further prevent lower respiratory tract infections for which the airways are blessed with several innate defense mechanisms among which mucociliary clearance is one of the most important ones (1).

It is a known and evident fact that the dynamic protective functioning of the entire respiratory system comes from the mucociliary apparatus that continuously removes harmful inhaled particles before they can reach the distal bronchioles and alveoli.

Respiratory cilia are microscopic motile projections present on the apical surface of pseudostratified columnar epithelial cells lining the nasal cavity, paranasal sinuses, trachea, bronchi, and larger bronchioles (2). Approximately 200 motile cilia are present on each ciliated epithelial cell that beat in a highly coordinated fashion to propel mucus toward the pharynx. This process, known as the mucociliary escalator, is the first line of defense against inhaled pathogens and toxins (3).

Multiple interacting factors like ultrastructure of the cilia, frequency rate of beating of the cilia, ciliary coordination, composition of the mucus and airway surface liquid hydration all affect normal mucociliary clearance (4). Disturbance of any of these components impairs mucus transport and predisposes individuals to recurrent respiratory infections, chronic inflammation, that ultimately leads to progressive lung damage.

Interestingly, mal functioning of respiratory cilia has been linked to have a role in both inherited and acquired respiratory disorders and has therefore attracted a lot of attention. (5) Disorders such as primary ciliary dyskinesia, cystic fibrosis, COPD, severe asthma, and post-viral

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airway diseases are increasingly linked to abnormalities in ciliary function and epithelial defense mechanisms (5,6). Furthermore, the occurrence of COVID-19 pandemic renewed scientific interest in mucociliary defense as viral infections seems to directly impair ciliary integrity and airway clearance. (15)

This review discusses the physiology of respiratory cilia, the mechanisms underlying mucociliary clearance, the clinical consequences of ciliary dysfunction, recent advances in diagnostics and therapeutics, and future directions in respiratory ciliary research.

Structure and Ultrastructure of Respiratory Cilia:

Respiratory motile cilia are highly specialized organelles approximately 5–7 μm in length and 0.2 μm in diameter (2). Their origin is from basal bodies that are attached within the apical cytoplasm of epithelial cells. Structurally, motile cilia possess the classical “9+2” axonemal arrangement composed of nine peripheral microtubular doublets, two central microtubules, inner and outer dynein arms, radial spokes and nexin links (6)

Dynein arms are specific in that they contain ATPase enzymes responsible for generating sliding movement between adjacent microtubules. Coordinated sliding forces are converted into bending motions with the help of nexin links while radial spoke help in the stabilization of the entire arrangement during rhythmic ciliary beating (7).

Ciliary orientation and its synchronization are maintained very efficiently by the basal body. Also, proper alignment of cilia allows coordinated metachronal beating, without which mucus transport cannot happen effectively. Any type of structural defects affecting dynein arms, radial spokes, central microtubules, or basal bodies will impair ciliary motion and contribute to ciliary dyskinesia syndromes (8).

Recent cryoelectron tomography studies performed by Gui et al (9) between 2020 and 2025 has contributed significantly in improving our understanding of ciliary ultrastructure and molecular architecture, mainly related to dynein assembly proteins and axonemal organization (9).

Physiology of Ciliary Beating:

Ciliary Beat Frequency: Normal ciliary beat frequency ranges between 7–20 Hz but vary by temperature, hydration, pH, and local signaling impulses re-

ceived (3). The ciliary beat cycle consists of two strokes: an effective stroke and recovery stroke

The effective stroke aims to propel mucus proximally and upwards by extending the cilia rigidly into the mucus layer where the recovery stroke largely focuses on minimizing the backward movement of mucus by bending the cilia closer to the epithelial surface.

Regulation of Ciliary Activity: Ciliary motility is regulated by complex intracellular signaling pathways that involves the interaction between calcium ions, cyclic AMP and GMP, nitric oxide and ATP-mediated purinergic signaling (3)

It has been seen that adrenergic stimulation generally works towards increasing ciliary beat frequency, whereas environmental toxins and inflammatory mediators may suppress ciliary activity. Also, nitric oxide plays an especially important role in modulating ciliary movement and maintaining epithelial homeostasis (10).

Recent research has shown that airway epithelial cells actively regulate ciliary function in response to environmental stimuli, inflammation, and infection. This dynamic regulation highlights the importance of cilia as active participants in airway immunity rather than passive mechanical structures.

Mucociliary Clearance Mechanism:

Not only is mucociliary clearance dependent on the working of the respiratory cilia but also on the coordinated interaction between respiratory cilia, airway mucus and airway surface liquid (ASL) (11). It is worth noting that the ASL consists of two layers, the mucus gel layer which traps inhaled particles, bacteria, allergens, and pollutants and the periciliary liquid layer a low-viscosity fluid that permits free ciliary movement. (11).

As the cilia beats in this ASL fluid and at the same time interacts with the mucus therefore proper hydration of the ASL is of utmost importance because dehydration will compress the periciliary layer and thereby restricts ciliary movement. This mechanism is particularly important in the pathophysiology of diseases such as cystic fibrosis, where defective ion transport causes ASL dehydration and impaired mucus clearance (12).

Computational and biophysical studies from 2020 onward have demonstrated that mucus viscosity, elasticity, and hydration significantly influence mucociliary transport efficiency (13). Excessively thick mucus impairs

ciliary propulsion, leading to mucus stasis and infection susceptibility.

Respiratory Cilia in Airway Defense:

One of the important components of innate pulmonary immunity is the respiratory cilia. Their major functions include not only removal of inhaled pathogens, clearance of toxic particles but also prevention of microbial colonization and maintenance of airway sterility that ultimately prevents or limits inflammation related injury.

By this process of continuous clearing of mucus and debris, cilia prevent pathogens from reaching distal lung regions (4). Efficient mucociliary clearance also minimizes prolonged exposure of epithelial cells to inflammatory mediators and environmental toxins. A study published by Fahy et al in 2021 showed that impaired mucociliary clearance is associated with increased risk of bacterial biofilm formation, chronic airway inflammation, and recurrent respiratory infections (14).

Respiratory viral infections have shown to damage ciliated epithelial cells directly where viruses such as influenza and SARS-CoV-2 cause ciliary loss, abnormal beating patterns, epithelial disruption, and impaired mucociliary transport, thereby increasing vulnerability to secondary bacterial infections (15).

Environmental Factors Affecting Respiratory Cilia:

Cigarette Smoke:

One of the most important acquired causes of ciliary dysfunction has been proven time and again to be Cigarette smoke. Tobacco smoke exposure affects the cilia in a number of ways with mainly reducing ciliary beat frequency and causing its shortening and also causing a loss of ciliated epithelial cells, not only this, it adversely causes goblet cell hyperplasia with increased mucus production (16).

Apart from this oxidative stress and inflammatory mediators generated by smoking also contribute in damaging both ciliary ultrastructure and epithelial integrity. Chronic impairment of mucociliary clearance contributes ultimately to COPD pathogenesis.

Air Pollution:

With air pollution on the rise, it is not surprising that particulate matter, industrial pollutants, and toxic gases

are seen to be linked towards impairing ciliary motility and epithelial health. Fine particulate pollutants penetrate deep into the airways and induce oxidative injury, inflammation, and mucus hypersecretion (17). Urban air pollution is increasingly recognized as an important contributor to chronic airway diseases through its effects on mucociliary defense mechanisms. (17)

Viral Infections:

Respiratory viruses directly act on and damage airway epithelial cells and impair mucociliary transport. Severe respiratory viral infection has shown to reduce ciliary density and disrupt ciliary coordination causing epithelial shedding and altering mucus composition (15). Because of this post-viral ciliary dysfunction may contribute to persistent respiratory symptoms and chronic airway inflammation.

Ciliary Dysfunction & Respiratory Diseases:

Primary Ciliary Dyskinesia (PCD): Primary ciliary dyskinesia is a rare inherited disorder affecting approximately 1 in 10,000–20,000 individuals with autosomal recessive inheritance patterns and is characterized by abnormal ciliary structure or motility resulting in defective mucociliary clearance (5,18).

Genetics of PCD involves more than 50 genes that control the structure and assembly of PCD (19,20). These genes encode directly for proteins involved in dynein arm formation, axonemal assembly, radial spoke stabilization and intraflagellar transport. Recent genotype-phenotype studies have demonstrated substantial clinical variability depending on the underlying mutation (20). Certain genetic defects are associated with more severe lung disease, earlier bronchiectasis, and faster decline in pulmonary function.

Pathophysiology of PCD involves defective ciliary motility that leads to impaired mucus clearance and chronic retention of secretions within the airways. Persistent mucus stasis promotes bacterial colonization, recurrent infections, and chronic neutrophilic inflammation (18). With time, chronic inflammation damages airway walls and causes progressive bronchiectasis. The frequent occurrence of infections further impairs lung growth and pulmonary function.

Longitudinal study of lung disease in primary ciliary dyskinesia by Davus et al. have shown that the early diagnosis and aggressive airway clearance therapy sig-

nificantly improve long-term outcomes in PCD patients (21).

Chronic Obstructive Pulmonary Disease (COPD):

COPD is associated with multiple abnormalities in mucociliary clearance including reduced ciliary density, ciliary shortening and impaired ciliary coordination along with excess mucus production and airway dehydration (16). All of these contribute together to increase susceptibility to respiratory infections and cause acute exacerbations.

Cystic Fibrosis: Mutations in the CFTR chloride channel lead to impaired chloride and water transport across epithelial surfaces resulting in cystic fibrosis (12). This results in dehydration of airway surface liquid that compresses the periciliary layer and severely impairs mucociliary clearance.

It is remarkable to note that although ciliary ultrastructure may remain relatively preserved, thick viscoelastic mucus prevents effective transport. Recent studies suggest that restoring airway hydration significantly improves ciliary efficiency and mucus clearance (22).

Asthma: Asthma is associated with alteration in both ciliary coordination and ciliary orientation where microscopic examination shows epithelial inflammation, mucus hypersecretion, and altered ciliary coordination. Moreover, abnormalities linked to epithelial structure affects ciliary orientation and function also seems to accompany severe asthma. Along with this, there is usually a rise in inflammatory cytokines such as IL-13 that promote goblet cell hyperplasia and mucus overproduction, which interfere with mucociliary transport (23).

Chronic Rhinosinusitis: Patients with chronic rhinosinusitis have also shown to exhibit impaired mucociliary clearance frequently owing to epithelial inflammation, stasis in mucus movement, and altered ciliary function. Unfortunately, persistent impairment contributes to recurrent infections and chronic sino nasal symptoms (24) and poor quality of life.

Diagnostic Advances in Ciliary Disorders:

High-Speed Video Microscopy: Direct visualization of ciliary beat pattern and frequency can be done using high-speed video microscopy so with time this technique has become an important diagnostic tool for PCD and acquired ciliary disorders (25).

Electron Microscopy: The use of electron microscopy permits the identification of ultrastructural abnormalities such as dynein arm defects, central pair abnormalities and microtubular disorganization. Despite it being a useful tool, yet the number of cases of functional ciliary disorders cannot be ignored where anatomical appearance is absolutely normal in appearance.

Nasal Nitric Oxide Measurement: Patients with PCD have shown to have markedly reduced nasal nitric oxide levels. Based on this, nasal nitric oxide testing is being used as a screening tool due to its high sensitivity (26).

Genetic Testing: Genetic studies have now proven to be diagnostic and also as a preventive measure where next-generation sequencing panels can identify causative mutations in the majority of PCD patients. Also, Genetic testing facilitates family counseling and prognostic assessment (20).

Artificial Intelligence-Assisted Diagnostics: With advancing technology especially in the field of medical devices, one of the promising tools available is artificial intelligence-based image analysis system that work towards automated assessment of ciliary beat frequency, beat coordination and ultrastructural abnormalities. These technologies may improve diagnostic accessibility and reduce observer variability (27).

Emerging Therapeutic Strategies:

Airway Clearance Therapy: Certain physical maneuvers and management focus largely on providing mucus clearance using techniques like chest physiotherapy, positive expiratory pressure devices, oscillatory devices etc. The role of regular exercise also cannot be overlooked here or be ignored as airway clearance remains central to management of PCD and bronchiectasis.

Hypertonic Saline and Hydration Therapy: Hypertonic saline improves airway hydration is a proven fact and because of that it enhances mucociliary transport by increasing periciliary liquid depth (22). Owing to that hydration therapy is regarded as an indispensable supportive strategy in chronic airway diseases associated with impaired mucus clearance.

Anti-inflammatory Therapies: Excessive neutrophilic inflammation has been directly linked to contribute significantly to airway damage in ciliary disorders. Keeping with this mind set many anti-inflammatory

therapies targeting epithelial cytokines and inflammatory mediators are also currently under investigation (23).

Gene Therapy: Experimental studies are ongoing for gene- editing approaches and some advances in CRISPR-Cas9 technology and mRNA therapeutics have brought along a large interest in gene correction strategies for inherited ciliary disorders (20) as they may eventually provide disease-modifying treatment for PCD.

Stem Cell and Regenerative Medicine: Already studies are being done involving airway epithelial stem cell therapies and organoid technologies for regeneration of damaged ciliated epithelium (28) as these methodologies can bring in future treatment benefits for patients with severe airway injury and inherited ciliary disease.

Future Directions:

From the future perspectives, the respiratory ciliary medicine needs to delve deeper into approaches involving modern technologies like Precision medicine, personalized genetic therapies, AI –assisted and nanotechnology-based diagnosis and therapy. At the same time, experts need to look into regenerative airway medicine and advance care for improvement of the disorders.

Also, integration of translational medicine, molecular genetics, and computational biology will surely lead to highly targeted therapies that may be capable of restoring normal mucociliary function. (29).

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